Equivariant Systems Theory and Observer Design for Autonomous Systems

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Necessary files

Download the following files from Dropbox:

- `Homography_Estimator_skel.m` - skeleton code for observer
- `homogWarp.m` - helper function to visualize homographies
- `imdisp.m` - helper function for image display
- `license_imdisp.txt` - license file for imdisp
- `DATASET.zip` - image and gyro data

Put all of the files into a single folder and unzip the dataset. Make sure that this creates a subfolder named `DATASET` with all the data files in it.

Edit Homography_Estimator_skel.m

```
% COMPLETE START: add the part of the observer dynamics
% that does NOT depend on the image point measurements
% Omega_x: angular velocity in so(3)
Hhat = ;
Gamma = ;
% COMPLETE END
```

[...]

```
% COMPLETE START: add the innovation terms in the observer
% kP: gain for Hhat
% kI: gain for Gamma
% pref: 3 x nbrPoints matrix of _normalized_ homogeneous coordinates of
% reference points, pref(:,j) is the j-th point
% pcur: 3 x nbrPoints matrix of _normalized_ homogeneous coordinates of
% points in current image, pcur(:,j) is the j-th point
% PointsValid(j): ==1 if jth point can be used, 0 otherwise
Hhat = ;
Gamma = ;
% COMPLETE END
```

Homography observer

This observer assumes that the linear velocity is constant *in the reference frame* (different to the case treated in the previous lecture).

$$e_i = \frac{\hat{H} p_i}{|\hat{H} p_i|}$$

$$\alpha = \sum_{i=1}^{n} (I - e_i e_i^\top) \mathring{p}_i e_i^\top$$

$$\dot{\hat{H}} = \hat{H}(\Omega_\times + \hat{\Gamma}) + k_P \alpha \hat{H},$$
$$\dot{\hat{\Gamma}} = [\hat{\Gamma}, \Omega_\times] + k_I \operatorname{Ad}_{\hat{H}^T} \alpha$$

$$[\hat{\Gamma}, \Omega_\times] = \hat{\Gamma}\Omega_\times - \Omega_\times \hat{\Gamma}$$
$$\operatorname{Ad}_{\hat{H}^T} \alpha = \hat{H}^T \alpha \left(\hat{H}^T\right)^{-1}$$